HCov Exam Sheet 2025/26 - The Gratwick Files

Theorem: Random Theorems

- **Triangle Inequality** $|z+w| \le |z|+|w|$
- **Reverse Triangle Inequality** $|z-w| \ge ||z|-|w||/$

Remark: Notation

Look at the context and these are usually obvious

- $z = x+iy = \Re(z)+i\Im(z)$
- $z_0 = x_0+iy_0, w = a+ib$
- $S, U, D \subseteq \mathbb{C}$
- $f(z) = u(x,y)+iv(x,y)$
- $(x,y) \in \mathbb{R}^2$
- $f: S \to C$, $D \to C$, etc (context :)))
- z, z_0, w etc $\in \mathbb{C}, S, D$ etc
- Γ is a contour/curve (context :)))
- $[a,b] \subseteq \mathbb{R}$
- 'Respects linearity' or 'is linear' for an operation means can do $\alpha f + \beta g$ and pull out sum + factors from the operation.
- (T/P/L)S/E = (Taylor/Power/Laurent) Series/Expansion
- If a disc/circle/etc is demarcated with a random R/r/etc, assume it is 'for some R/r/etc >0'

Complex Fundamentals

Definition: Basics

- $\mathbb{C} = \{x+iy \mid x,y \in \mathbb{R}\}$ is a field
- $\Re(z) = \frac{z+\overline{z}}{2}$, $\Im(z) = \frac{z-\overline{z}}{2i}$
- $z = re^{i\theta}$: $x = r\cos\theta$, $y = r\sin\theta$
- $e^{in\theta} = (e^{i\theta})^n$
- $|z| = 0 \iff z = 0$
- $\left|z^2\right| = |z|^2 = z\overline{z}$
- $\overline{zw} = (\overline{z})(\overline{w})$
- $0! = 1$, $f^{(0)} = f$
- $r \downarrow 0$ indicates $r \to 0$ through positive values only

Definition: The Argument

- For some $z = re^{i\theta}$, θ is an *argument* of z. $\mathrm{Arg}(z)$ is the unique $\theta \in (-\pi, \pi]$
- $\arg(z) = \{\theta : z = |z|e^{i\theta}\} = \{\mathrm{Arg}(z) + 2k\pi : k \in \mathbb{Z}\}$
- $\arg(zw) = \arg(z) + \arg(w)$, $\arg(\overline{z}) = -\arg(z)$
- $\mathrm{Arg}(zw) = \mathrm{Arg}(z) + \mathrm{Arg}(w) \pm 2k\pi$, $\mathrm{Arg}(\overline{z}) = -\mathrm{Arg}(z) \pm 2k\pi$
- $\mathrm{Arg}_\phi(z)$ is the branch of $\arg(z)$ s.t $\phi < \mathrm{Arg}_\phi(z) \le \phi + 2\pi$

Theorem: Properties Of Sets $S \subseteq \mathbb{C}$

- *Open* if $\forall\ z_0 \in U$, $\exists \epsilon > 0$ s.t. $D_\epsilon(z_0) \subseteq U$. For example, $D_\epsilon(z), D'_\epsilon(z)$.
- *Closed* if its complement is open, and iff $S = \overline{S}$. For example $\overline{D}_\epsilon(z)$
- A *neighbourhood* of z_0 is an open set containing it.
- z_0 is a *limit point* of S if $\forall \epsilon > 0, D'_\epsilon(z_0) \cap S$ is nonempty
- $\overline{S}$, the *closure* of S, is the union of S with its limit points.
- *Bounded* if $\exists M > 0$ s.t. $|z| \le M$, for all $z \in S$. *Unbounded* otherwise.

Definition: Complex Sequences

Complex sequences $z_n \in \mathbb{C}$ are analogous to real ones, except index from 0.

- $z = \lim_{n\to\infty} z_n$ if $\forall \epsilon > 0$, there exists $N \in \mathbb{N}$ s.t. $|z_n - z| < \epsilon$ for $n \ge N$.
- z_n can be split into the combination of real sequences, $z_n = a_n + ib_n$. $z = \lim_{n\to\infty} z_n$ iff $\Re(z) = \lim_{n\to\infty} a_n$ and $\Im(z) = \lim_{n\to\infty} b_n$. z_n converges iff a_n and b_n converge.
- Cauchy, Bolzano-Weierstrass, boundedness are all akin to $\mathbb{R}$
- $z \in \overline{S}$ iff there exists a sequence $z_n \in S$ s.t. $z = \lim_{n\to\infty} z_n$

Definition: Complex Functions

$f(z)$ can be written as $f(x+iy) = u(x,y) + iv(x,y)$, where $u,v : \mathbb{R}^2 \to \mathbb{R}$

- For $z_0 \in \overline{S}$, $a_0 = \lim_{z\to z_0} f(z)$ if $\forall\ \epsilon > 0$, $\exists \delta > 0$ s.t. $|f(z) - a_0| < \epsilon$ when $0 < |z - z_0| < \delta$. Standard $\epsilon - \delta$ definition.
- The lim exists iff the lims of the $\Re/\Im$ exist and are the $\Re/\Im$ of the limit
- Let $z_0 \in \overline{S}, f : S \to \mathbb{C}$, $\lim_{z\to z_0} f(z) = a_0$, and $w_n \in S \backslash \{z_0\}$ s.t $w_n \to z_0$, then $\lim_{n\to\infty} f(w_n) = a_0$.
- Limit algebra is analogous to the reals

Definition: Continuity

- $f(z)$ is cont. at z_0 if $\lim_{z\to z_0} f(z)$ exists and equals $f(z_0)$, and cont. on $S \subseteq \mathbb{C}$ if it is cont. at all $z \in S$.
- $f(z)$ is cont. at z_0 iff u,v are cont. at (x_0, y_0).
- $+/\times/\div$(where denominator $\neq 0$) of cont. funcs. preserves continuity.
- f is cont. iff the preimage $f^{-1}(U)$ is open $\forall$ open $U \subseteq \mathbb{C}$
- Continuous functions map clounded sets to clounded sets

Definition: Differentiability and Holomorphicity

- $f(z)$ diffable (and thus cont.) at z_0 if $\lim_{z\to z_0} \frac{f(z) - f(z_0)}{z - z_0}$ exists.
- $f(z)$ is holo. at z_0 if it is diffable on a neighbourhood of z_0
- $+/\cdot/\div$(where denom $\neq 0$) of holo. funcs. preserves holo.city
- Composition preserves diffability, $(f \circ g)'(z_0) = f'(g(z_0))g'(z_0)$

Major Theorem: The Cauchy-Riemann Equations

The *Cauchy-Riemann equations* state that if $f(z)$ is diffable at z_0 then $\frac{\partial u}{\partial x}(x_0,y_0) = \frac{\partial v}{\partial y}(x_0,y_0)$ and $\frac{\partial u}{\partial y}(x_0,y_0) = -\frac{\partial v}{\partial x}(x_0,y_0)$

- If u,v continuously diffable (derivative is continuous) on a neighbourhood of (x_0,y_0), and satisfy the C-R equations at (x_0,y_0), then f is diffable at z_0.
- C-R can be abbrev.ed by defining $\partial = \left(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\right)$, $\overline{\partial} = \left(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\right)$. Then f satisfies the C-R equations iff $\overline{\partial} f = 0$.

Definition: Harmonic Functions

$h : \mathbb{R}^2 \to \mathbb{R}$ is *harmonic* if $\forall (x,y) \in \mathbb{R}^2$: $\frac{\partial^2 h}{\partial x^2} + \frac{\partial^2 h}{\partial y^2} = 0$ (Laplace).

- Holomorphicity implies harmonicity
- For a harmonic u, v is a *harmonic conjugate of* u if $f = u + iv$ is holomorphic

Definition: Polynomials and Rational Functions

- For polynom $P(z) = \sum_{n=0}^{N} a_n z^n$, coeffs $a_i \in \mathbb{R}$, $P(z_0) = 0$ iff $P(\overline{z_0}) = 0$
- For polynoms P, Q, $R := \frac{P(z)}{Q(z)} : \{z \in \mathbb{C} : Q(z) \neq 0\} \to \mathbb{C}$ is a *rational function*.
- The domain of a rational function is always an open set
- Rational functions are holomorphic on $\{z \in \mathbb{C} : Q(z) \neq 0\}$

Definition: exp(z)and Complex Trig

- $\exp(z) := e^x(\cos(y) + i\sin(y))$
- $\exp(z) = \exp'(z)$ holo. on $\mathbb{C}$
- $\exp(z + 2\pi i) = \exp(z)$, so not bijective
- $\sin(z) = \frac{\exp(iz) - \exp(-iz)}{2i}$
- $\cos(z) = \frac{\exp(iz) + \exp(-iz)}{2}$
- $\sin'(z) = \cos(z)$, $\cos'(z) = -\sin(z)$
- $\cos^2(z) + \sin^2(z) = 1$
- $\tan(z) := \frac{\sin(z)}{\cos(z)}$, $\cot(z) := \frac{1}{\tan(z)}$
- $\sec(z) := \frac{1}{\cos(z)}$, $\csc(z) = \frac{1}{\sin(z)}$
- $\sinh(z) := \frac{\exp(z) - \exp(-z)}{2}$
- $\cosh(z) := \frac{\exp(z) + \exp(-z)}{2}$
- Periodicity, $\frac{\pi}{2}$ offset akin to $\mathbb{R}$

- $\sin(x+iy) = \sin(x)\cosh(y) + i\cos(x)\sinh(y)$
- $\cos(x+iy) = \cos(x)\cosh(y) - i\sin(x)\sinh(y)$
- $\sin(z+w) = \sin(z)\cos(w) + \cos(z)\sin(w)$
- $\cos(z+w) = \cos(z)\cos(w) - \sin(z)\sin(w)$
- $\sinh(iz) = i\sin(z)$, $\cosh(iz) = \cos(z)$

Definition: The Complex Logarithm

$\log(z) := \{w \in \mathbb{C} : \exp(w) = z\}$
$\mathrm{Log}(z) := \{\ln|z| + i\phi : \phi \in \arg(z)\} = \ln|z| + i\mathrm{Arg}(z)$

- $\log(re^{i\theta}) = \ln|z| + i\arg(z) = \{\ln r + i\theta + 2k\pi i : k \in \mathbb{Z}\}$
- $\log(zw) = \log(z) + \log(w)$, $\log(\frac{1}{z}) = -\log(z)$
- $\mathrm{Log}_\phi(z) := \ln|z| + i\mathrm{Arg}_\phi(z)$ holo. on D_ϕ, $\mathrm{Log}'_\phi(z) = \frac{1}{z}$ for all $z \in D_\phi$

Definition: Branches and Cuts

Motivated by $\mathrm{Arg}(z)$'s discontinuity on the negative real line, which means $\mathrm{Log}(z)$ is not holo. on all of $\mathbb{C}$.

- A branch cut is a line $L \subseteq \mathbb{C}$ which is such that that a holomorphic branch of a multivalued func may be defined on $\mathbb{C} \backslash L$
- $L_{z_0,\phi} := \{z \in \mathbb{C} : z = z_0 + re^{i\phi}$ for $r > 0\}$. $L_{0,\phi} = L_\phi$.
- $D_{z_0,\phi} = \mathbb{C} \backslash L_{z_0,\phi}$. $D_{0,\phi} = D_\phi$. $D_{-\pi} = D$.

Definition: Complex Powers

$z^\alpha := \{\exp(\alpha w) : w \in \log(z)\} = \{\exp(\alpha \mathrm{Log}(z))\exp(i\alpha 2k\pi) : k \in \mathbb{Z}\}$

- Let $\alpha, z \in \mathbb{C}, z \neq 0$:
  - (i) If $\alpha \in \mathbb{Z}$ z^α has exactly one value.
  - (ii) If $\alpha = \frac{p}{q}$, $p,q \in \mathbb{Z}, p,q$ coprime, $q \neq 0$, z^α has exactly q values.
  - (iii) If α is irrational or non-real, z^α has infinitely many values.
- Let $q \in \mathbb{Z}^+$. $1^{\frac{1}{q}} = 1, w, \ldots, w^{q-1}$, $w := \exp(\frac{2\pi i}{q})$ are the q *roots of unity*
- Each branch of $\log(z)$ gives a branch of z^α.
- The *principal branch* is $z^\alpha = \exp(\alpha \mathrm{Log}(z))$
- For $z \neq 0$, $z^\alpha z^\beta = z^{\alpha+\beta}$ choosing the principal branch for each power.
- A branch of z^α is holomorphic on the cut plane D_ϕ on which Log_ϕ is holomorphic, and $\forall z \in D_\phi$, $\frac{d}{dz} z^\alpha = \alpha z^{\alpha-1}$.

Conformal Maps and Möbius Transforms

Definition: Conformal Map

For open $U \subseteq \mathbb{C}, f : U \to \mathbb{C}$ is *conformal* if the angle between straight lines is equal to the angle between their images under f. This extends to diffable curves by considering their tangents.

- A holomorphic $f(z)$ is conformal for all $z_0 \in U$ where $f'(z_0) \neq 0$

Definition: The Extended Complex Plane

The *extended complex plane* is $\tilde{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$.

Let $a, b \in \mathbb{C}, b \neq 0$, then $a + \infty = \infty$, $b \cdot \infty = \infty$, $\frac{b}{0} = \infty$, $\frac{b}{\infty} = 0$.

Then $f(-\frac{d}{c}) = \infty$ and $f(\infty) = \frac{a}{c}$. We do not define $0 \cdot \infty$ or $\infty + \infty$.

Definition: Möbius Transforms

A *Möbius transformation* is a func of form $f(z) = \frac{az+b}{cz+d}, a,b,c,d \in \mathbb{C}, ad \neq bc$

- $\frac{\lambda az + \lambda b}{\lambda cz + \lambda d} = \frac{az+b}{cz+d}$, can always set $\lambda = \sqrt{(ad-bc)}$ and impose $ad - bc = 1$
- $f_M(z) = \frac{az+b}{cz+d}$ corresponds to $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $\det(M) = ad - bc = 1$.

Translation	*Rotation*	*Dilation*	*Inversion*
$f(z) = z + b$	$f(z) = e^{i\theta} z$	$f(z) = rz, r > 0$	$f(z) = \frac{1}{z}$
$\begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}$	$\begin{pmatrix} e^{i\theta/2} & 0 \\ 0 & e^{-i\theta/2} \end{pmatrix}$	$\begin{pmatrix} \sqrt{r} & 0 \\ 0 & 1/\sqrt{r} \end{pmatrix}$	$\begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$

Theorem: Möbius Transform Properties

Let $f(z)$ be a Möbius Transform and $z_2, z_3, z_4 \in \tilde{C}$ be distinct. Then:

- Every MT is a composition of a finite number of translations, rotations, and dilations, and iff f does not fix the point at infinity, one inversion.
- MTs map circlines to circlines

- $f_{M_1 M_2} = f_{M_1} \circ f_{M_2}$ and $f_{M^{-1}} = f_M^{-1}$
- If $f(z_2) = z_2, f(z_3) = z_3, f(z_4) = z_4$, $f = id$.
- For There exists a unique f s.t. $f(z_2) = 1, f(z_3) = 0, f(z_4) = \infty$)
- For triplets of distinct points $(z_2, z_3, z_4), (w_2, w_3, w_4) \in \tilde{\mathbb{C}}$, $\exists$ a unique MT f s.t. $f(z_2) = w_2, f(z_3) = w_3, f(z_4) = w_4$
- f preserves the cross-ratio: $[f(z_1), f(z_2), f(z_3), f(z_4)] = [z_1, z_2, z_3, z_4]$

Definition: The Riemann Sphere and Stereographic Projection

Describe $\mathbb{R}^3$ with $\{X, Y, Z\}$. $\mathbb{C}$ is the plane $Z = 0$ and $X + iY$ is $\{X, Y, 0\}$:

- The *Riemann Sphere* is $S^2 := \{(X,Y,X) \in \mathbb{R}^3 : X^2 + Y^2 + Z^2 = 1\}$.
- The 'North Pole' is the point $N := \{0, 0, 1\}$
- $\phi : \tilde{\mathbb{C}} \to S^2$ is defined by: $\phi(z), z, N$ are collinear, and $\phi(\infty) = N$.
- The *stereographic projection* is $\psi := \phi^{-1}(z) : S^2 \to \mathbb{C}$.
- ψ maps circles to circlines.
- $\phi(z) = (\frac{2x}{|z|^2+1}, \frac{2y}{|z|^2+1}, \frac{|z|^2-1}{|z|^2+1})$. $\psi(X,Y,Z) = \frac{X+iY}{1-Z}$, $\psi(N) = \infty$.

Major Definition: The Cross-Ratio

For distinct $z_1, z_2, z_3, z_4 \in \tilde{\mathbb{C}}$, their *cross-ratio*, $[z_1, z_2, z_3, z_4]$, is the image of z_1 under the unique f s.t. $f(z_2) = 1, f(z_3) = 0, f(z_4) = \infty$. $[z_1, z_2, z_3, z_4] = \frac{z_1 - z_3}{z_1 - z_4} \frac{z_2 - z_4}{z_2 - z_3}$.

$[\infty, z_2, z_3, z_4] = \frac{z_2 - z_4}{z_2 - z_3}$, $[z_1, \infty, z_3, z_4] = \frac{z_1 - z_3}{z_1 - z_4}$,

$[z_1, z_2, \infty, z_4] = \frac{z_2 - z_4}{z_1 - z_4}$, $[z_1, z_2, z_3, \infty] = \frac{z_1 - z_3}{z_2 - z_3}$.

Complex Integration

Definition: Complex Integration Of A Real Variable

For $[a,b] \subseteq \mathbb{R}$, $f : [a,b] \to \mathbb{C}$ is *integrable* if $u,v : [a,b] \to \mathbb{R}$ are integrable. Then $\int_a^b f(t)\ \mathrm{d}t = \int_a^b u(t)\ \mathrm{d}t + i\int_a^b v(t)\ \mathrm{d}t$. Continuous functions are integrable. This integration respects linearity, FTC, and $\left|\int_a^b f(t)\ \mathrm{d}t\right| \le \int_b^a |f(t)|\ \mathrm{d}t$

Definition: Contours And Their Types

For $z_0, z_1 \in \mathbb{C}$, a *(parametrised) curve* Γ connecting them is a continuous func $\gamma : [t_0, t_1] \to \mathbb{C}$ for $t_0, t_1 \in \mathbb{R}, t_0 < t_1$, s.t. $\gamma(t_0) = z_0$ and $\gamma(t_1) = z_1$.

- γ can be split into $\Re/\Im$ parts by two continuous real functions $x, y : [t_0, t_1] \to \mathbb{R}$ s.t. $\gamma(t) = x(t) + iy(t)$ for all $t \in [t_0, t_1]$
- Γ is *regular* if γ is continuously diffable and $\gamma'(t) \neq 0$ for all $t \in (t_0, t_1)$
- Γ is a closed and bounded subset of $\mathbb{C}$
- For regular Γ, *arclength* $\ell(\Gamma) := \int_{t_0}^{t_1} \left|\gamma'(t)\right|\ \mathrm{d}t = \int_{t_0}^{t_1} \sqrt{x'(t)^2 + y'(t)^2}\ \mathrm{d}t$
- For Γ an arc of a circle of radius r through an angle θ, $\ell(\Gamma) = r\theta$
- A curve Γ is a *contour* if it is a finite union of regular curves $\Gamma_1, \ldots, \Gamma_N$ s.t $\gamma_n(t_1) = \gamma_{n+1}(t_0)$, ie the endpoints join up.
- A contour is *simple* if it has no self-intersections, except possibly at t_0/t_1.
- A *loop* is a simple closed contour
- A loop Γ is *positively-oriented* if when moving around Γ, $\mathrm{Int}(\Gamma)$ is on the left.

Definition: Contour Integrals

Let $z_0, z_1 \in \mathbb{C}$, Γ be a contour from z_0 to z_1, and $f : \Gamma \to \mathbb{C}$ be continuous.

Then the *integral of f along* Γ is defined by $\int_\Gamma f(z)\ \mathrm{d}z = \int_{t_0}^{t_1} f(\gamma(t))\gamma'(t)\ \mathrm{d}t$

- Contour integration is linear
- For two parametrisations of Γ, $\gamma : [t_0, t_1] \to \mathbb{C}$ and $\tilde{\gamma} : [\tilde{t_0}, \tilde{t_1}] \to \mathbb{C}$, if there exists injective diffable $\lambda : [\tilde{t_0}, \tilde{t_1}] \to [t_0, t_1]$ with $\lambda'(t) > 0 \forall t \infty [\tilde{t_0}, \tilde{t_1}]$ s.t. $\tilde{\gamma}(t) = \gamma(\lambda(t))\ \forall\ t \in [t_0, t_1]$, then the integrals along γ and $\tilde{\gamma}$ are equal.
- If Γ is parametrised by $\gamma : [0,1] \to \mathbb{C}$, then integrating $-\Gamma$ parametrised by $\tilde{\gamma} : [0,1] \to \mathbb{C}, \tilde{\gamma}(t) = \gamma(1-t)$ gives the negative of the integral along Γ.
- A contour is *closed* if $\gamma(t_0) = \gamma(t_1)$
- A *contour integral* of f is the sum of the integrals of f along the regular curves which make up the contour.

Definition: Domains

$D \subseteq \mathbb{C}$ is a *domain* if D is open and every two points in D can be connected by a contour lying wholly in D.

- If $u : D \to \mathbb{R}$ is diffable, with $\frac{\partial u}{\partial x} = \frac{\partial u}{\partial y} = 0$ on D, then u is constant on D.
- A cont. f on D has an *antiderivative* F on D if $F'(z) = f(z)$ for all $z \in D$.
- The integral of cont. f with an antideriv on D along a closed Γ in D is 0.
- D is *simply-connected* if $\mathrm{Int}(\Gamma) \subseteq D$ for all loops $\Gamma \subseteq D$.

Lemma: The M-L Lemma

Let Γ be a regular curve in $\mathbb{C}$ and $f : \Gamma \to \mathbb{C}$ be continuous. Then $\left|\int_\Gamma f(z)\ \mathrm{d}z\right| \le \max_{z \in \Gamma} |f(z)| \ell(\Gamma)$

Lemma: Path-Independence

For a domain $D \subseteq C$ and cont. $f : D \to \mathbb{C}$, the following are equivalent:

- (i) f has an antiderivative F on D
- (ii) $\int_\Gamma f(z)\ \mathrm{d}z = 0$ for all closed Γ in D
- (iii) All contour integrals are independent of the path Γ, and depend only on the endpoints

Theorem: Jordan Curve Theorem

Let Γ be a loop in $\mathbb{C}$. Then Γ partitions $\mathbb{C}$ into two domains: one bounded, the *interior* of Γ, $\mathrm{Int}(\Gamma)$; and one unbounded, the *exterior* of Γ, $\mathrm{Ext}(\Gamma)$.

- $\mathbb{C} = \mathrm{Int}(\Gamma) \cup \Gamma \cup \mathrm{Ext}(\Gamma)$

Major Theorem: Cauchy Integral Theorem

Let Γ be a loop and f be holo. inside and on Γ. Then $\int_\Gamma f(z)\ \mathrm{d}z = 0$.

- If f is holo. on a simply-connected domain D, f has an antideriv on D
- If $z_0 \in \mathbb{C}$ and $z_0 \notin \Gamma$, then $\int_\Gamma \frac{1}{z - z_0}\ \mathrm{d}z = 2\pi i$ if $z \in \mathrm{Int}(\Gamma)$ and 0 otherwise

Theorem: The Deformation Theorem

Let $\Gamma_1, \Gamma_2 \subseteq \mathbb{C}$ be loops and f holo. on: $\Gamma_1; \Gamma_2; \{\mathrm{Int}(\Gamma_1) \backslash \mathrm{Int}(\Gamma_2)\} \cup \{\mathrm{Int}(\Gamma_2) \backslash \mathrm{Int}(\Gamma_1)\}$. Then $\int_{\Gamma_1} f(z)\ \mathrm{d}z = \int_{\Gamma_2} f(z)\ \mathrm{d}z$

Major Theorem: Cauchy Integral Formula, Generalised CIF

Let Γ be a loop, $z_0 \in \mathrm{Int}(\Gamma)$, and f holo. inside and on Γ.

Then: For loop Γ, , f holo. inside and on Γ: $f(z_0) = \frac{1}{2\pi i} \int_\Gamma \frac{f(z)}{z - z_0}\ \mathrm{d}z$

Given a loop Γ, $z_0 \in \mathrm{Int}(\Gamma)$, and f holo. inside and on Γ: f is infinitely differentiable at z_0 and for $n \in \mathbb{N}$, $f^{(n)}(z_0) = \frac{n!}{2\pi i} \int_\Gamma \frac{f(z)}{(z - z_0)^{n+1}}\ \mathrm{d}z$

- Can be rearranged to calculate contour integrals if appropriate derivatives are known
- For $z_0 \in$ domain D, $R > 0$ s.t. $\overline{D}_R(z_0) \subseteq D$, f holo. on D, and $M > 0$ an upper bound for $|f(z)|$, then for all $n \in \mathbb{N}$, $\left|f^{(n)}(z_0)\right| \le \frac{n! M}{R^n}$

Theorem: Morera's Theorem

If $D \subseteq \mathbb{C}$ a domain, $f : D \to \mathbb{C}$ cont., $\int_\Gamma f(z)\ \mathrm{d}z = 0$ for all loops Γ inside D, then f is holo.

Theorem: Liouville's Theorem

If f is holo. on $\mathbb{C}$ and bounded (ie $\exists M > 0$ s.t. $|f(z)| \le M \forall z \in \mathbb{C}$), then f is constant.

- For $D \subseteq \mathbb{C}$ a domain, $z_0 \in D$, $R > 0$ s.t. $\overline{D}_R(z_0) \subseteq D$, and f holo. on D; $f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + R\exp(it))\ \mathrm{d}t$.
- Given the above and f s.t. $\max_{z \in \overline{D}_R(z_0)} |f(z)| = |f(z_0)|$, $f(z)$ is constant on $\overline{D}_R(z_0)$.

Theorem: The Maximum Modulus Principle

Let $D \subseteq \mathbb{C}$ be a domain, and f be holo. and bounded on D. Then if $|f(z)|$ achieves its maximum on D, f is constant on D.

- If a func is holo. on a bounded domain, and cont. up to and including the boundary, it attains the maximum of its modulus on the boundary.

Thm: Maximum/Minimum Principle For Harmonic Funcs

Let $D \subseteq \mathbb{R}^2$ be a domain and $\phi : D \to \mathbb{R}$ be harmonic s.t. ϕ is bounded above or below on D by $M > 0$, and $\phi(z_0) = M$ for some $z_0 \in D$. Then ϕ is constant on D.

Series Expansions For Holomorphic Functions

Definition: Convergence Of An Infinite Series

An infinite series $\sum_{j=0}^{\infty} z_j$ *converges* if the sequence $S_n \in \mathbb{C}, S_n = \sum_{j=0}^{n} z_j$ is convergent with limit $S \in \mathbb{C}$, and say $\sum_{j=0}^{\infty} z_j = S$. Otherwise the series is *divergent*.

- If $\sum_{j=0}^{\infty} z_j$ is convergent, $\lim n \to \infty z_n = 0$
- Comparison Test: If z_n is s.t. $|z_n| \le M_n \ge 0$, for all $n \ge N$ for some $N \in \mathbb{N}$, and $\sum_{j=0}^{\infty} M_j$ converges, then $\sum_{j=0}^{\infty} z_j$ converges.
- Geometric Series: For $c \in \mathbb{C}$, $\sum_{j=0}^{\infty} c^j$ converges (to $\frac{1}{1-c}$) iff $|c| < 1$.
- Ratio Test: If $\lim_{n\to\infty} \left| \frac{z_{n+1}}{z_n} \right| = L$, $\sum_{j=0}^{\infty} z_j$ converges if $L < 1$ and diverges if $L > 1$. If $L = 1$ the test gives no information..

Definition: Pointwise And Uniform Convergence

For $S \subseteq \mathbb{C}$ and $f_n : S \to \mathbb{C}$ a sequence of functions:
f_n converge *pointwise* to $f : S \to \mathbb{C}$ if for all $z \in S$ and all $\epsilon > 0$, there exists $N \in \mathbb{N}$ s.t. $|f_n(z) - f(z)| < \epsilon$ when $n \ge N$. Note N may depend on z.
f_n converge *uniformly* to $f : S \to \mathbb{C}$ if for all $\epsilon > 0$, there exists $N \in \mathbb{N}$ s.t. $|f_n(z) - f(z)| < \epsilon$ when $n \ge N$. Note N may not depend on z.

- Holo. f_n may converge pointwise to an f which is not even cont.
- If cont. f_n converge uniformly to f, f is cont.
- $\sum_{j=0}^{\infty} f_j(z)$ converges pointwise/uniformly if $S_n = \sum_{j=0}^{n} f_j$ converges pointwise/uniformly.
- Weierstrass M-Test: For $S \subseteq C, M_n \ge 0$ s.t. $\forall z \in S$ and $n \ge N \in \mathbb{N}$, $|f_n(z)| \le M_n$, and $\sum_{j=0}^{\infty} M_j$ convergent; $\sum_{j=0}^{\infty} f_j(z)$ is uniformly convergent.
- For $f_n : S \to \mathbb{C}$ uniformly converging to f, and $\Gamma \subseteq S$; $\int_\Gamma f_n(z)\ \mathrm{d}z \to \int_\Gamma f(z)\ \mathrm{d}z$.
- For cont. f_n s.t. $\sum_{j=0}^{\infty} f_j(z)$ converges uniformly on S and $\Gamma \subseteq S$, $\int_\Gamma \sum_{j=0}^{\infty} f_j(z)\ \mathrm{d}z = \sum_{j=0}^{\infty} \int_\Gamma f_j(z)\ \mathrm{d}z$
- For simply-connected D with f_n holo. on D converging uniformly to f, f is holomorphic on D.

Definition: Power Series

For $a_n \in \mathbb{C}$ a sequence, a *power series* is an infinite series of form $\sum_{j=0}^{\infty} a_j (z - z_0)^j$. a_n are the *coefficients*.
Given a power series, $\exists R \in [0, \infty) \cup \{\infty\}$, the *radius of convergence*, such that:

(i) The series converges on $D_R(z_0)$
(ii) The series converges uniformly on $\overline{D}_r(z_0)$ for $r \in [0, R]$
(iii) The series diverges on $\mathbb{C} \backslash \overline{D}_R(z_0)$

- For $f = \sum_{j=0}^{\infty} a_j (z - z_0)^j$, f is holo. on $D_R(z_0)$
- If $\lim_{j\to\infty} \left| \frac{a_j}{a_{j+1}} \right|$ exists, it is equal to R.

Definition: Taylor Series

For f holo. at $z_0 \in \mathbb{C}$,
the *Taylor series of f centred at z_0* is the PS $\sum_{j=0}^{\infty} \frac{f^{(j)}(z_0)}{j!} (z - z_0)^j$.

- For f holo. on $D_R(z_0)$, the TS of f centred at z_0 converges to $f(z)$ for all $z \in D_R(z_0)$, uniformly on $\overline{D}_r(z_0)$ for $0 \le r < R$
- Thus the TS of f centred at z_0 converges to f everywhere inside the largest open disc centred at z_0 on which f is holo.
- For U open, f is *analytic* if for all $z \in U$, f can be expressed as a convergent power series centred at z, valid on some disc centred at z.
- If U is open and f holo. , f is analytic
- For f holo. on and $z \in D_R(z_0)$, $f'(z) = \sum_{j=0}^{\infty} \frac{f^{(j+1)}(z_0)}{j!} (z - z_0)^j$
- For f, g holo. on and $z \in D_R(z_0)$, TSs respect linearity and the TS for fg, valid on same centre/disc, is $\sum_{j=0}^{\infty} \frac{1}{j!} \left(\sum_{k=0}^{j} \binom{j}{k} f^{(k)}(z_0) g^{(j-k)}(z_0) \right) (z - z_0)^j$
- Uniqueness: If $\exists$ a PS converging to f on a disc, it is the TS of f.

Definition: Laurent Series

For $z_0 \in \mathbb{C}$ and $\ldots, a_{-1}, a_0, a_1, \ldots \in C$
a doubly-infinite sequence, a *Laurent series* centred at z_0 is a series of form $\sum_{j=-\infty}^{\infty} a_j (z - z_0)^j = \sum_{j=0}^{\infty} a_j (z - z_0)^j + \sum_{j=1}^{\infty} a_{-j} (z - z_0)^{-j}$

- An LS converges if the two component PSs converge. The first has RoC R, regard the second as a PS in variable $w = (z - z_0)^{-1}$ with RoC $S > 0$, then the LS converges for z s.t. $|z - z_0| < R$ and $\left| (z - z_0)^{-1} \right| < S$, or $S^{-1} = r < |z - z_0| < R$.
- For f holo. on $A_{r,R}(z_0), 0 \le r < R \le \infty$, f can be expressed an an LS centred at z_0 that converges on $A_{r,R}(z_0)$, uniformly on $\overline{A}_{r',R'}(z_0), r < r' \le R' < R$, with coeffs $a_j = \frac{1}{2\pi i} \int_\Gamma \frac{f(z)}{(z - z_0)^{j+1}}\ \mathrm{d}z$, for Γ in $A_{r,R}(z_0)$ s.t. $z_0 \in \mathrm{Int}(\Gamma)$
- If f is holomorphic on the 'hole' of the annulus, the LS is the TS
- Uniqueness: If the PS $\sum_{j=-\infty}^{\infty} c_j (z - z_0)^j$ converges on $A_{r,R}(z_0)$, $\exists f$ holo. on $A_{r,R}(z_0)$ s.t. its LS centred at z_0 is the PS. Ie, if a func can be expressed in terms of a PS of the form of an LS centred at some z_0, that PS is the LS

Definition: zeroes

Let $z_0 \in \mathbb{C}$,
U a neighbourhood of z_0, f holo. on U z_0 is a *zero* of f if $f(z_0) = 0$.
z_0 is of *finite order* if there exists $m \in \mathbb{N}$ s.t. $f(z_0) = f'(z_0) = \ldots = f^{(m-1)}(z_0) = 0$ but $f^m(z_0) \ne 0$. Then z_0 is a *zero of order m*.

- A zero of order 1 is a *simple zero*.
- A zero is *isolated* if $\exists R > 0$ s.t. $f(z) \ne 0$ for $z \in D'_R(z_0)$
- If f has a zero of finite order at z_0, then it is isolated
- If f is such that $f(z_n) = 0$ for a sequence of distinct $z_n \in U$ s.t. $z_n \to z_0$ then f is identically zero on some disc centred at z_0. (Proof by Taylor expansion since zero at z_0 is of infinite order, and non-isolated)

Definition: Singularities

Let D be a domain, $z_0 \in \mathbb{C}$, $f : D \to \mathbb{C}$ z_0 is a *singularity* of f if f is not holo. at z_0.
If z_0 is an isolated singularity of f holo. on $D'_R(z_0)$ for some $R > 0$, then f has an LE $f(z) = \sum_{j=-\infty}^{\infty} a_j (z - z_0)^j$ valid on $A_{0,R}$. Then:

(i) z_0 is a *removable singularity* of f if $a_j = 0$ for all $j < 0$, ie $f(z) = \sum_{j=0}^{\infty} a_j (z - z_0)^j$
(ii) z_0 is a *pole of order m* of f, $m \in \mathbb{N}$, if $a_j = 0$ for $j < -m$ and $a_{-m} \ne 0$, ie $f(z) = \sum_{j=-m}^{\infty} a_j (z - z_0)^j$. Poles of order 1,2,3 are *simple, double, triple* poles.
(iii) z_0 is an *essential singularity* if $a_j \ne 0$ for infinitely many negative values of j

If z_0 is a removable singularity of f holo. on $D'_R(z_0)$ for $R > 0$, then $f(z_0)$ can be redefined s.t. f is holo. at z_0. By definition f has an LE valid on $D'_R(z_0)$, so setting $f(z_0) = a_0$, the value of the LE at z_0, makes it holo.

- A singularity is *isolated* if there exists $R > 0$ s.t. f is holo. on $D'_R(z_0)$.
- If z_0 is a singularity of a rational function $f = \frac{P}{Q}$, it is isolated
- For f, g holo. at z_0, with z_0 a zero of order m of g:
  (i) If z_0 is not a zero of f, then f/g has a pole of order m at z_0
  (ii) If z_0 is a zero of order k of f, then f/g has a pole of order $m - k$ at z_0 if $m > k$, and a removable singularity at z_0 otherwise.

Definition: Analytic Continuation

For $D \subseteq \tilde{D} \subseteq \mathbb{C}$ domains, and $f : D \to \mathbb{C}$ holo., we say a holo. func $F : \tilde{D} \to \mathbb{C}$ is an *analytic continuation* of f if $F(z) = f(z)$ for $z \in D$.

Major Theorem: The Identity Theorem

Let $D \subseteq \mathbb{C}$ be a domain, $z_0 \in D$, f holo. on D, and $f(z) = 0$ for all $z \in D_R(z_0)$. Then $f(z) = 0$ for all $z \in D$.

- If $g : D \to \mathbb{C}$ is also holo. and s.t. $f(z) = g(z)$ for all $z \in D_R(z_0)$, then $f(z) = g(z)$ for all $z \in D$.
- If f is s.t. $f(z_n) = 0$ for a seq. of distinct $z_n \in D, z_n \to z_0$, then $f(z) = 0$ for all $z \in D$.
- If $g(z)$ also s.t. $g(z_n) = 0$ for the same z_n, then $f(z) = g(z)$ for all $z \in D$.

Residue Calculus And Its Applications

Theorem: 5.1.1

Let $z_0 \in \mathbb{C}$, f be holo. on $D'_R(z_0)$ with an isolated sing. at z_0, and Γ a loop inside $D'_R(z_0)$ with $z_0 \in \mathrm{Int}(\Gamma)$. Then $\int_\Gamma f(z)\ \mathrm{d}z = 2\pi i a_{-1}$, where a_{-1} is the $j = -1$ coeff in the LE of f, centred at z_0 valid on $D'_R(z_0)$.

Definition: The Residue

Let $z_0 \in \mathbb{C}$, f holo. on $D'_R(z_0)$ with an isolated singularity at z_0. Then the *residue of f at z_0* is $\mathrm{Res}(f, z_0) = a_{-1}$ where the LS of f valid on $D'_R(z_0)$ is $f(z) = \sum_{j=-\infty}^{\infty} a_j (z - z_0)^j$

Lemma: Residue Calculation Lemmas

For all, let $z_0 \in \mathbb{C}$ and f holo. on $D'_R(z_0)$

- 5.1.4: If f has a removable singularity at z_0, $\mathrm{Res}(f, z_0) = 0$.
- 5.1.5: If f has a pole of order m at z_0 then $\mathrm{Res}((f, z_0) = \lim_{z \to z_0} \frac{1}{(m-1)!} \frac{d^{m-1}}{dz^{m-1}} (z - z_0)^m f(z$
- 5.1.7: If $f = g/h$, g, h holo. on $D_R(z_0)$ s.t. h has a simple zero at z_0 and $g(z_0) \ne 0$, then $\mathrm{Res}(f, z_0) = \frac{g(z_0)}{h'(z_0)}$

Major Theorem: The Cauchy Residue Theorem

Let Γ be a loop, and f holo. inside and on Γ except for finitely many isolated singularities $z_1, \ldots, z_k \in \mathrm{Int}(\Gamma)$. Then $\int_\Gamma f(z)\ \mathrm{d}z = 2\pi i \sum_{j=1}^{k} \mathrm{Res}(f, z_j)$

Definition: Meromorphicity

Let $D \subseteq \mathbb{C}$ be a domain. Then: f is *meromorphic on D* if for all $z \in D$, f has a pole of finite order at z or f is holo. at z.

- If Γ a loop in D, f mero. on D and not identically 0, then f has a finite number of poles and zeroes on $\mathrm{Int}(\Gamma)$.

Definition: $N_0(f)$ and $N_\infty(f)$

Let Γ be a loop and f mero. on $\mathrm{Int}(\Gamma)$, with zeroes $w_1, \ldots, w_l$ and poles $z_1, \ldots, z_k$ all $\in \mathrm{Int}(\Gamma)$. Then $N_0(f)$, the number of zeroes of f inside Γ, and $N_\infty(f)$, the number of poles of f inside Γ, both counted with multiplicity, are: $N_0(f) = \sum_{j=1}^{l}$ order of w_j, and $n_\infty(f) = \sum_{j=1}^{k}$ order of z_j.

Theorem: The Argument Principle

Let Γ be a loop in $\mathbb{C}$, f be mero. on $\mathrm{Int}(\Gamma)$, and holo. and non-zero on Γ. Then: $\int_\Gamma \frac{f'(z)}{f(z)}\ \mathrm{d}z = 2\pi i (N_0(f) - N_\infty(z))$.

- If f is holo. inside and on Γ, then $\int_\Gamma \frac{f'(z)}{f(z)}\ \mathrm{d}z = 2\pi i (N_0(f))$

Theorem: Rouché's Theorem

Let Γ be a loop and f, g holo. on and inside Γ s.t. for all $z \in \Gamma$, $|f(z) - g(z)| < |f(z)|$. Then $N_0(f) = N_0(g)$.

- FTA: A polynomial of degree n has exactly n zeros counted with multiplicity

Major Theorem: Open Mapping Theorem

Let $D \subseteq \mathbb{C}$ be a domain, and f non-constant and holo. on D. Then the image of D under f, $f(D) = \{f(z) : z \in D\}$ is an open subset of $\mathbb{C}$

- Let f be holo. on D s.t. any of $\Re(f(z)), \Im(f(z)), |f(z)|, \mathrm{Arg}(f(z))$ are constant. Then f is constant.
- Schwarz Lemma: Let f be holo. on $D_1(0)$ s.t. $f(0) = 0$ and $|f(z)| \le 1$ for all $z \in D_1(0)$. Then $|f(z)| \le |z|$ for all $z \in D_1(0)$.

Definition: Improper Integrals

Let $f : \mathbb{R} \to \mathbb{R}$ be continuous. Then

(i) $\int_0^\infty f(x)\ \mathrm{d}x := \lim_{R\to\infty} \int_0^R f(x)\ \mathrm{d}x$
(ii) $\int_{-\infty}^0 f(x)\ \mathrm{d}x := \lim_{R\to-\infty} \int_R^0 f(x)\ \mathrm{d}x$
(iii) $\int_{-\infty}^\infty f(x)\ \mathrm{d}x := \lim_{R_1, R_2 \to (\infty, -\infty)} \int_{R_2}^{R_1} f(x)\ \mathrm{d}x$

whenever the appropriate limits exist. If $\int_{-\infty}^\infty f(x)\ \mathrm{d}x$ exists, it is equal to $\lim_{R\to\infty} \int_R^R f(x)\ \mathrm{d}x$.
Let $c \in (a, b) \subseteq \mathbb{R}$ and $f : [a, b] \backslash \{c\} \to \mathbb{R}$ be cont.. Then:

(i) $\int_a^c f(x)\ \mathrm{d}x := \lim_{r\downarrow 0} \int_a^{c-r} f(x)\ \mathrm{d}x$
(ii) $\int_c^b f(x)\ \mathrm{d}x := \lim_{s\downarrow 0} \int_{c+s}^b f(x)\ \mathrm{d}x$
(iii) $\int_a^b f(x)\ \mathrm{d}x := \lim_{r\downarrow 0} \int_a^{c-r} f(x)\ \mathrm{d}x + \lim_{s\downarrow 0} \int_{c+s}^b f(x)\ \mathrm{d}x$

whenever the appropriate limits exist. If the limits in the third case exist the result is equal to the p.v. of the integral.

Definition: Principal Value

For $c \in \mathbb{R}$, $f : \mathbb{R} \backslash \{c\} \to \mathbb{R}$ cont., define the *principal value* of the integral by: $\mathrm{p.v.} \int_{-\infty}^\infty f(x)\ \mathrm{d}x = \lim_{R\to\infty, r\downarrow 0} \left(\int_{-r}^{c-r} f(x)\ \mathrm{d}x + \int_{c+r}^R f(x)\ \mathrm{d}x \right)$ provided the two limits exist independently

- If $\lim_{R\to\infty} \int_R^R f(x)\ \mathrm{d}x$ exists and $\int_{-\infty}^\infty f(x)\ \mathrm{d}x$ does not, define the *Cauchy principal value* by $\mathrm{p.v.} \int_{-\infty}^\infty f(x)\ \mathrm{d}x := \lim_{R\to\infty} \int_R^R f(x)\ \mathrm{d}x$.

Lemma: Jordan Lemma

Let P, Q be polynoms with $\deg(Q) \ge \deg(P) + 1$. Consider $R := P/Q$ and let $a \in \mathbb{R} \ne 0$. Then: $\lim_{R\to\infty} \int_{C_R^+} \exp(iaz) \frac{P(z)}{Q(z)}\ \mathrm{d}z = 0$ if $a > 0$ and $\lim_{R\to\infty} \int_{C_R^-} \exp(iaz) \frac{P(z)}{Q(z)}\ \mathrm{d}z = 0$ if $a < 0$.

Lemma: 5.5.3 -Integrating On An Arc

Let $D \subseteq$ be a domain, $c \in D$, f mero. on D with a simple pole at $z \in D$, and S_r be the circular arc paremed. by $\gamma(\theta) = c + r\exp(i\theta)$ for $\theta \in [\theta_0, \theta_1], 0 \le \theta_0 < \theta_1 \le 2\pi$. Then $\lim_{r\downarrow 0} \int_{S_r} f(z)\ \mathrm{d}z = i(\theta_1 - \theta_0)\mathrm{Res}(f, c)$

Lemma: 5.6.5 - The Binomial As As Integral

Let $0 \le k \le n$ be non-negative integers, $\binom{n}{k} = \frac{n!}{k!(n-k!)}$, and Γ a loop with $0 \in \mathrm{Int}(\Gamma)$. Then: $\binom{n}{k} = \frac{1}{2\pi i} \int_\Gamma \frac{(1+z)^n}{z^{k+1}}\ \mathrm{d}z$.

Examples

Example: Trigonometric Integrals

Integrals of form $\int_0^{2\pi} R(\cos\theta, \sin\theta)\ \mathrm{d}\theta$ for rational R can be evaluated by considering a contour integral of an appropriate function around $C_1(0)$.
Recall for $z = e^{i\theta}$ on $C_1(0)$,
$\cos\theta - \Re(z) = \frac{z + \overline{z}}{2} = \frac{1}{2}\left(z + \frac{1}{z}\right)$ and $\sin\theta - \Im(z) = \frac{z - \overline{z}}{2i} = \frac{1}{2i}\left(z - \frac{1}{z}\right)$
Define
$f(z) = \frac{1}{iz} R\left(\frac{z + 1/z}{2}, \frac{z - 1/z}{2i}\right)$, then $f(\exp(i\theta)) = \frac{1}{i\exp(i\theta)} R(\cos\theta, \sin\theta)$.
Parametrise $C_1(0)$ by $\gamma : [0, 2\pi] \to \mathbb{C}, \gamma(\theta) = \exp(i\theta)$.
Then $\int_{C_1(0)} f(z)\ \mathrm{d}z = \int_0^{2\pi} f(\gamma(\theta))\gamma'(\theta)\ \mathrm{d}\theta = \int_0^{2\pi} f(\exp(i\theta)) i\exp(i\theta)\ \mathrm{d}\theta = \int_0^{2\pi} \frac{1}{i\exp(i\theta)} R(\cos\theta, \sin\theta) i\exp(i\theta)\ \mathrm{d}\theta = \int_0^{2\pi} R(\cos\theta, \sin\theta)\ \mathrm{d}\theta$. So we may transform a real integral to a contour integral, then evaluate using the CRT.

Example: Evaluating Infinite Series With Contour Integrals